Journal of *Marine Science and Engineering*

MDPI

Article

A CDC–ANFIS-Based Model for Assessing Ship Collision Risk in Autonomous Navigation

Hee-Jin Lee [1] and Ho Namgung [2,*]

[1] Division of Education and Planning, Korea Institute of Maritime and Fisheries Technology, Busan 49111, Republic of Korea; heejin1911@naver.com
[2] Department of Maritime Industry Convergence, Mokpo National Maritime University, Mokpo 58628, Republic of Korea
* Correspondence: ngh2009@mmu.ac.kr

Abstract

To improve collision risk prediction in high-traffic coastal waters and support real-time decision-making in maritime navigation, this study proposes a regional collision risk prediction system integrating the Computed Distance at Collision (CDC) method with an Adaptive Neuro-Fuzzy Inference System (ANFIS). Unlike Distance at Closest Point of Approach (DCPA), which depends on the position of Global Positioning System (GPS) antennas, Computed Distance at Collision (CDC) directly reflects the actual hull shape and potential collision point. This enables a more realistic assessment of collision risk by accounting for the hull geometry and boundary conditions specific to different ship types. The system was designed and validated using ship motion simulations involving bulk and container ships across varying speeds and crossing angles. The CDC method was used to define collision, almost-collision, and near-collision situations based on geometric and hydrodynamic criteria. Subsequently, the FIS–CDC model was constructed using the ANFIS by learning patterns in collision time and distance under each condition. A total of four input variables—ship speed, crossing angle, remaining time, and remaining distance—were used to infer the collision risk index (CRI), allowing for a more nuanced and vessel-specific assessment than traditional CPA-based indicators. Simulation results show that the time to collision decreases with higher speeds and increases with wider crossing angles. The bulk carrier exhibited a wider collision-prone angle range and a greater sensitivity to speed changes than the container ship, highlighting differences in maneuverability and risk response. The proposed system demonstrated real-time applicability and accurate risk differentiation across scenarios. This research contributes to enhancing situational awareness and proactive risk mitigation in Maritime Autonomous Surface Ship (MASS) and Vessel Traffic System (VTS) environments. Future work will focus on real-time CDC optimization and extending the model to accommodate diverse ship types and encounter geometries.

Keywords: ship collision; fuzzy inference system; CDC; MASS; CRI

check for updates

Academic Editor: Cheng Liu

Received: 7 July 2025
Revised: 31 July 2025
Accepted: 1 August 2025
Published: 1 August 2025

Citation: Lee, H.-J.; Namgung, H. A CDC–ANFIS-Based Model for Assessing Ship Collision Risk in Autonomous Navigation. *J. Mar. Sci. Eng.* **2025**, *13*, 1492. https://doi.org/10.3390/jmse13081492

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

1. Introduction

Ship collisions at sea can lead to severe consequences, including casualties, property damage, and environmental pollution [1]. According to statistics from the Korean Maritime Safety Tribunal [2], collision accidents have accounted for a high proportion of maritime accidents over the past five years and have frequently led to secondary incidents [3–5]. Due

to these issues, continuous efforts have been made in the field of maritime safety to prevent collision accidents.

The advent of the era of maritime autonomous surface ships (MASSs) is anticipated to usher in a new maritime traffic paradigm. Consequently, many researchers are striving to develop new collision avoidance systems [6–9]. For the safe operation of MASSs, a systematic approach will be essential, one that can quantitatively assess collision risk and determine appropriate evasive actions in various sea conditions [10,11]. The development of a collision risk assessment model capable of actively responding to real-time changes in the surrounding environment while adhering to the international maritime organization (IMO)'s International Regulations for Preventing Collisions at Sea (COLREGs, 1972) is emerging as a particularly crucial task [12,13].

Therefore, this study aims to complement the structural limitations of the CRI and CDC. The objective is to propose a precise and realistic model applicable to MASSs, based on the CDC concept, but by subdividing the existing CRI framework by Namgung and Kim [14] according to specific ship type characteristics and through actual collision scenarios. The experimental subjects were primarily selected as container ships and bulk ships, which have distinct differences in maneuvering performance [4]. Based on this, simulations were conducted using CDC, and these results were used to improve the structure of the CIR's collision risk index. Furthermore, this study intends to demonstrate that a more realistic collision assessment is possible by incorporating interaction effects between ships, which were not considered in the existing CDC.

Literature Review

In this context, the collision risk index (CRI) has garnered significant attention as a crucial component in the decision-making systems of autonomous ships [15–22].

Hasegawa et al. [17] conducted one of the pioneering studies utilizing expert systems, proposing the Ship Auto-navigation Fuzzy Expert System (SAFES). This research aimed to formalize the knowledge and experience of skilled mariners into a rule-base of "If-Then" statements. By combining these rules with fuzzy logic, the system was designed to respond to uncertain maritime situations. The SAFES received navigational data such as the position and speed of surrounding vessels, used fuzzy inference to assess the degree of risk, and proposed avoidance maneuvers based on the COLREGs. This work was a seminal attempt to model the complex decision-making process of a human navigator within a computer system. Lee and Rhee [18] advanced the field by integrating an expert system with a search algorithm. They recognized that the qualitative avoidance rules provided by an expert system alone were insufficient for determining the optimal route. To address this, they introduced a search algorithm like A*. Their system employed a two-stage approach. First, the expert system generated several COLREG-compliant avoidance alternatives. Then, the search algorithm was applied to each alternative to calculate the optimal avoidance path, comprehensively considering factors such as time, fuel consumption, and safety. This method combined the intuitive nature of a rule-based system with the optimization capabilities of a search algorithm, enabling more efficient and safer avoidance maneuvers. Ahn et al. [19] moved beyond traditional expert systems by presenting an approach that fused artificial neural networks (ANNs) with fuzzy logic. In this study, an ANN was trained on actual ship navigation or simulation data to build a model capable of predicting collision risk. The neural network allowed the system to learn the non-linear and complex interactions between multiple vessels from data. The risk level and navigational parameters output by the network were then used as inputs for a fuzzy logic system to determine the final avoidance action. This marked a shift away from fixed, pre-defined rules toward a system that could improve its situational judgment capabilities through data-driven

learning. Bukhari et al. [20] shifted the research perspective from the individual ship to the Vessel Traffic System (VTS) center. They developed an intelligent system for assessing multi-vessel collision risk in real time from the viewpoint of a VTS operator. The core of this system was to use various navigational variables, including the Variation in Compass Degree (VCD), as inputs into a fuzzy inference system. This system calculated the risk level for each vessel pair and synthesized this information to visualize the overall traffic risk within the control area. The focus of this research was less on the maneuvering of an individual ship and more on empowering VTS operators to identify high-risk vessel groups early and perform proactive traffic management. Ohn and Namgung [21] further enhanced the uncertainty-handling capabilities of fuzzy logic by applying an Interval Type-2 Fuzzy Inference System to ship collision avoidance. Arguing that conventional Type-1 fuzzy systems have limitations in managing linguistic ambiguity and data uncertainty, they used a Type-2 system to model these uncertainties more effectively. They specifically applied this system to risk assessment based on the Closest Point of Approach (CPA), demonstrating that it could make more robust and reliable collision risk judgments and avoidance decisions by accounting for factors like measurement errors and the unpredictability of the marine environment. This study significantly advanced the practical applicability of such systems in real-world maritime settings by improving their reliability. Namgung and Kim [14] developed a CRI model by formalizing collision avoidance maneuvers according to COLREGs and proposing appropriate response distances for each avoidance stage. This model dynamically adjusts a ship's domain based on its speed and size, calculating the collision risk index using various navigational factors such as Distance to Closest Point of Approach (DCPA), Time to Closest Point of Approach (TCPA), VCD, and relative distance (DR). Notably, they also utilized an Adaptive Neuro-Fuzzy Inference System (ANFIS) to train their system and developed FIS-NC (fuzzy inference system for near-collision), a fuzzy-inference-based decision-making system. The ANFIS combines the rule-based approach of a fuzzy inference system (FIS) with the learning capabilities of artificial neural networks [22]. This model can assess collision probability in non-linear maritime situations through data-driven learning [14]. FIS-NC is designed to provide decision criteria based on real-time situations, using the results learned from the ANFIS, making it directly applicable to the navigational judgments of MASSs.

However, this model has the following limitations. First, it relies excessively on the ship domain concept, meaning that risk assessment results can vary depending on how the domain is set. Second, it does not sufficiently reflect the specific characteristics of different ship types, which can lead to reduced accuracy in collision situations between dissimilar ships like container ships and bulk ships. Third, indicators such as DCPA and TCPA are calculated based on GPS antennas, but actual collisions occur at locations like the bow or stern, which can lead to inconsistencies with the realistic collision point [2].

To overcome the limitations of the aforementioned CPA-based collision assessment, Lee and Furukawa [3] proposed the concept of Computed Distance at Collision (CDC). CDC calculates the distance between two ships at all angles where a collision is possible, considering their actual shapes and motion characteristics. This method allows for a more precise collision assessment than existing DCPA-based models. Notably, CDC is differentiated by its ability to determine actual ship collisions without relying on the domain concept.

Traditional ship collision risk models have primarily relied on proximity-based indicators such as the Closest Point of Approach (CPA), Time to CPA (TCPA), and variations in ship domains—often in conjunction with expert rules derived from COLREGs [3,4].

However, these models often had limitations in fully accounting for dynamic ship maneuverability in real-world scenarios. Addressing these limitations, Montewka et al. [23]

introduced a novel approach to geometric collision probability estimation based on maritime and aviation experience, integrating actual vessel traffic data and generalized vessel dynamics through advanced statistical and optimization methods like Monte Carlo and genetic algorithms. Separately, focusing on enhancing the detection of critical navigational situations, Szłapczyński and Niksa-Rynkiewicz [24] developed a method for automatic near-miss detection based on the ship domain concept, relative speed, and course difference, utilizing a Mamdani neuro-fuzzy classification system. Additionally, Tam and Bucknall [25] contributed to encounter situation analysis by proposing a model that classifies the target ship's position and encounter type based on the heading angles of both the system's own ship and the target ship using a six-area division around the system's own ship for precise, COLREG-compliant distinctions between vessels requiring avoidance.

In contrast to these approaches, our study introduces a fundamentally different methodology that leverages CDC, which directly incorporates ship hull geometry, interaction angles, and kinematic profiles to estimate realistic collision states. By fusing CDC with the ANFIS, the proposed FIS–CDC model enables a data-driven inference of collision risk across vessel types and encounter geometries—offering improved interpretability and potential real-time applicability in autonomous ship operations and VTS environments.

However, CDC also has limitations. First, while it allows for precise calculations by reflecting a ship's shape and characteristics, applying this universally to all ship types presents significant practical constraints. Second, interaction effects between ships and errors due to simulation-based research can occur. In other words, while CDC is effective as a precise analysis tool for imminent collisions, it has limitations when used as a real-time, responsive system for systemic collision risk assessment, which is necessary for autonomous ships.

2. Methodology

In this section, the execution of the simulation will be detailed. This involves specifically outlining the model ship's exact specifications and the method for calculating its trajectory. CDC and the ANFIS will be introduced together with the precise approach to their integration within the study. Furthermore, the validation of the simulation environment and scenarios will be included to ensure the robustness and reliability of the results.

2.1. Configuration of the Scenario

Following the approach of Lee and Furukawa [3], this study assumes that a collision may occur when two ships are in close proximity within a confined waterway. Accordingly, it is essential to construct scenarios that identify regions where the give-way ship alone cannot prevent a collision due to the proximity of the encounter. To this end, the analysis focuses exclusively on crossing situations in which the crossing angle between the two ships (θ_{ot}) ranges from 2° to 90°, as illustrated in Figure 1.

In Figure 1, the horizontal and vertical axes denote the x and y coordinates, respectively. The black and red ellipses represent the own ship (OS) and the target ship (TS), respectively, while the red curved dotted line illustrates the initial trajectory of the TS. θ_{ot} denotes the relative heading angle between the two ships.

The maximum value of θ_{ot} is limited to 90°, as this represents the most critical crossing scenario in a narrow waterway, where the give-way ship may be required to undertake an evasive maneuver involving a 90° heading change to avoid a collision with the stand-on ship, as shown in Figure 2.

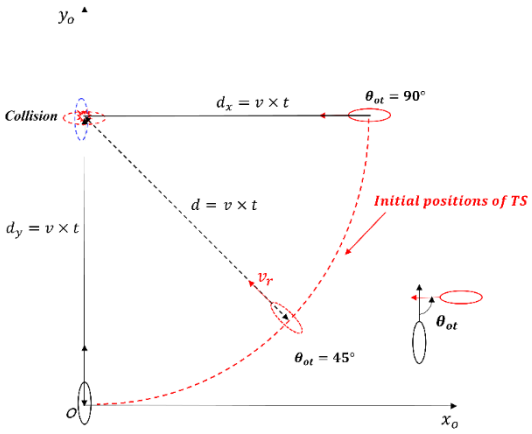


Figure 1. Configuration of the scenario.

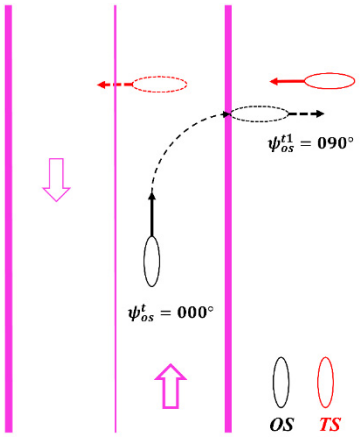


Figure 2. Scenario diagram.

In the scenario presented, the OS is designated as the give-way ship, while the TS serves as the stand-on ship. As illustrated in Figure 1, only the give-way ship employs rudder control to perform collision avoidance maneuvers. The rudder angle is used to compute the ship's trajectory, with a maximum angle of 35° applied to simulate emergency maneuvering.

For consistency, both ships are assumed to sail at the same speed and are modeled with identical dimensions. The ship models used in this study are bulk and container ships, which differ in the block coefficient and maneuvering characteristics. Given that container ships typically exhibit superior turning performance compared with bulk ships [4], the outcomes of this study are expected to differ from those reported previously. Additionally, in this study, the simulation was conducted without applying sea conditions. The ship speeds were set to 12, 14, and 16 kts for bulk carriers, considering their typical navigation speeds, and 16, 18, and 20 kts for container ships. The simulation calculations were performed at 0.1 s intervals.

2.2. Equations of Ship Motion

The distance between the two ships was calculated based on their respective coordinates, which were derived from the equations of ship motion. These equations require mathematical models that account for the hydrodynamic forces acting on the ship. In this study, the Maneuvering Modeling Group (MMG) model was adopted due to its proven accuracy and its capacity to separately consider the hydrodynamic forces generated by the hull, propeller, and rudder [26]. According to [27], the MMG model outperforms other maneu-

vering models, particularly in capturing the dynamic behavior of the propeller and rudder under realistic operating conditions, making it well-suited for course-keeping control.

To simulate the maneuvering motions of the ships, a model ship is required. In this study, bulk and container ship models were employed for both the OS and TS, with their principal particulars summarized in Table 1.

Table 1. Ship particulars.

Ship Parameter	Bulk Ship	Container Ship
Length, L (m)	233	274
Breadth, B (m)	32.30	32.30
Draft, d (m)	13.80	12.70
Block coefficient, C_b	0.88	0.65

Figure 3 illustrates both the Earth-fixed and body-fixed coordinate systems. Hydrodynamic forces acting on a ship are more accurately and effectively described within the body-fixed coordinate system [28]. The equations governing the ship's motion are expressed as follows:

$$\left.\begin{aligned} (m' + m'_x)\left(\frac{L}{U}\right)\left(\frac{\dot{U}}{U}cos\beta - \dot{\beta}sin\beta\right) + (m' + m'_y)r'^{sin\beta} &= X', \\ -(m' + m'_y)\left(\frac{L}{U}\right)\left(\frac{\dot{U}}{U}sin\beta - \dot{\beta}cos\beta\right) + (m' + m'_x)r'^{cos\beta} &= Y', \\ (I'_{zz} + i'_{zz})\left(\frac{L}{U}\right)^2\left(\frac{\dot{U}}{L}r' + \frac{U}{L}r'\right) &= N', \end{aligned}\right\} \quad (1)$$

where G denotes the center of gravity of the ship, while X and Y represent the hydrodynamic forces acting along the x and y axes, respectively. N denotes the moment about the vertical axis passing through G. The ship's velocity U is decomposed into components u and v along the x and y axes, respectively. ψ is the heading angle, β is the drift angle, and δ is the rudder angle.

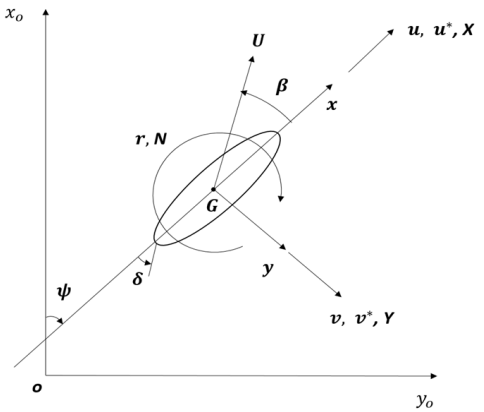


Figure 3. Earth- and body-fixed coordinate systems.

The prime symbol ($\prime$) indicates nondimensionalized quantities, and the dot above ($\dot{}$) denotes time differentiation. m', m'_x, and m'_y represent the mass and x- and y-axis components of the added mass of the ship, respectively. I'_{zz} and i'_{zz} correspond to the

nondimensional moment of inertia and added moment of inertia about the vertical axis. X', Y', and N' can be formulated as follows:

$$\left.\begin{array}{l} X' = X'_H + X'_P + X'_R, \\ Y' = Y'_H + Y'_R, \\ N' = N'_H + N'_R, \end{array}\right\} \tag{2}$$

where H, P, and R denote the contributions from the hull, propeller, and rudder, respectively.

The mathematical formulation of the nondimensional hydrodynamic forces and moment acting on the hull is based on the model proposed by [27].

$$\left.\begin{array}{c} X'_H = X'_{\beta r} r \prime sin\beta + X'_{uu} cos^2\beta \\ Y'_H = Y'_\beta \beta + Y'_r r' + Y'_{\beta\beta}\beta|\beta| + Y'_{rr} r'^{|r'|} + \left(Y'_{\beta\beta r}\beta + Y'_{\beta rr} r'\right)\beta r', \\ N'_H = N'_\beta \beta + N'_r r' + N'_{\beta\beta}\beta|\beta| + N'_{rr} r'^{|r'|} + \left(N'_{\beta\beta r}\beta + N'_{\beta rr} r'\right)\beta r', \end{array}\right\} \tag{3}$$

where $X'_{\beta r}$ indicates how the longitudinal force varies with respect to the drift angle β and nondimensional yaw rate r', X'_{uu} represents the hydrodynamic resistance encountered during straight-ahead motion, and Y'_β, Y'_r, N'_β, and N'_r represent linear hydrodynamic derivatives.

The propeller-induced longitudinal force X'_P is given by

$$\left.\begin{array}{c} X'_P = (1-t_P) n^2 D_P^4 K_T(J_P) / \frac{1}{2} L d U^2, \\ K_T(J_P) = C_1 + C_2 J_P + C_3 J_P^2, \\ J_p = U cos\beta (1 - w_P) / n D_P, \end{array}\right\} \tag{4}$$

where t_P is the thrust deduction coefficient, n denotes the propeller's rotational speed (in revolutions per second), D_P is the propeller diameter, $K_T(J_P)$ is the thrust coefficient as a function of the advance coefficient J_P, and C_1, C_2, and C_3 are empirical constants. w_P represents the effective wake fraction at the propeller location.

The hydrodynamic forces and moment induced by the rudder can be expressed as follows:

$$\left.\begin{array}{c} X'_R = -(1-t_R) F'_N sin\delta, \\ Y'_R = -(1+a_H) F'_N cos\delta, \\ N'_R = -(x'_R + a_H x'_H) F'_N cos\delta, \end{array}\right\} \tag{5}$$

where F'_N denotes the nondimensional rudder normal force, t_R and a_H are the interaction coefficients, and x'_R and x'_H represent the nondimensional positions of the rudder and the effective point of application of the additional lateral force generated by steering, respectively.

The above equations were employed to compute the trajectory of a model ship by numerically solving the system of differential equations presented in Equation (1).

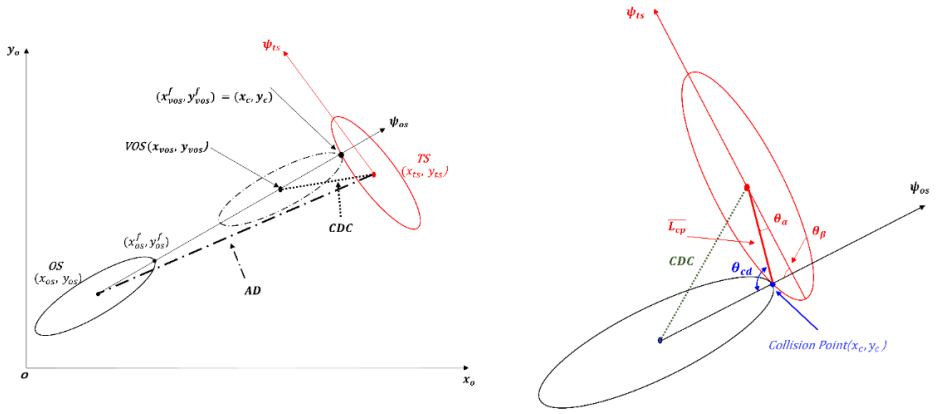
2.3. CDC

CDC, as proposed by Lee and Furukawa [3], is a ship CA model that calculates the collision distance at all angles at which two ships may collide. It compares this distance with the actual distance between the ships to assess whether and when a collision is likely to occur.

The accuracy of the model has been validated in previous studies [3–5]. The calculation procedure of CDC is as follows.

The black ellipse is the OS, the black dotted ellipse is the virtual OS (VOS), and the red ellipse is the TS in Figure 4a. In an Earth-fixed coordinate system, the reference point of the OS is denoted by x_{os}, y_{os}, the reference point of the TS is denoted by x_{ts}, y_{ts}, and their headings are denoted by ψ_{os} and ψ_{ts}, respectively. The reference point of the VOS

is denoted by x_{vos}, y_{vos}, and the coordinates of the OS and VOS headings are denoted by $x^f_{os}, y^f_{os}, x^f_{vos}$, and y^f_{vos} as shown in Figure 4a.



(**a**) (**b**)

Figure 4. Virtual OS positioning and concept of CDC. (**a**) Introduction of the VOS, (**b**) calculation of the CDC.

In Figure 4b, the collision point between the two ships is denoted by x_c, y_c. The CDC is defined as the distance between the two ships at the moment of collision, while AD represents the actual distance between them.

To compute the CDC, the OS is translated to a virtual position without altering its heading, such that it comes into contact with the TS. At this point, the heading direction of the OS becomes tangential to the boundary of the TS, meaning that the extended centerline of the OS is tangential to the TS's elliptical contour. The ellipse representing the TS is defined by the following equation:

$$\frac{\{(y_o - y_{ts})\cos\psi_{ts} + (x_o - x_{ts})\sin\psi_{ts}\}^2}{(L/2)^2} + \frac{\{-(y_o - y_{ts})\sin\psi_{ts} + (x_o - x_{ts})\cos\psi_{ts}\}^2}{(B/2)^2} = 1 \quad (6)$$

The OS's extended centerline is defined by:

$$y_o = (x_o - x_{os})\tan\psi_{os} + y_{os} \quad (7)$$

Applying Equation (7) to Equation (6) leads to the following:

$$Ax_o^2 + 2Bx_o + C = 0 \quad (8)$$

Accordingly, x_o can be determined.

$$x_o = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A} \; (x_o \in x_c) \quad (9)$$

The following terms are subsequently defined:

$$\left.\begin{aligned} L_{cp} &= \sqrt{(x_c - x_{ts})^2 + (y_c - y_{ts})^2}, \\ \theta_{cd} &= \psi_{os} + (2\pi - \psi_{ts}) + \tan^{-1}\left(\frac{x_c - x_{ts}}{y_c - y_{ts}}\right). \end{aligned}\right\} \quad (10)$$

Using these terms, the CDC can be computed as follows:

$$CDC = \sqrt{\left(\frac{L_{OS}}{2}\right)^2 + (L_{cp})^2 - 2\left(\frac{L_{OS}}{2}\right)L_{cp}\cos\theta_{cd}} \quad (11)$$

Using the reference positions of the OS and TS, the AD is determined as follows:

$$AD = \sqrt{(x_{os} - x_{ts})^2 + (y_{os} - y_{ts})^2} \tag{12}$$

In this study, we aimed to conduct a comprehensive collision risk assessment that includes not only simple physical collisions but also almost-collision (AC) and near-collision (NC) situations. The rationale for this is that, in extremely close encounters between ships, the risk of collision is considered to persist even without actual physical contact (e.g., at passing distances of 10–20 m) due to the potential for trajectory changes caused by hydrodynamic interactions and various uncertainties and errors in the prediction and measurement processes.

To this end, we referred to studies on ship-to-ship interactions [29–31]. The presented studies provide results from hydrodynamic experiments on ship interactions that can occur when two ships are in close proximity. We considered these results when determining the AC and NC values in this study. Of course, these values have a limitation in that they can vary depending on the size and type of the ship. An AC was defined as a minimum passing distance between two ships within 0.2 L, and an NC was defined as a passing distance within 0.4 L, which were applied in the assessment. While these standard values are not absolute, this study applied these definitions as an initial attempt to improve the evaluation criteria of CDC.

2.4. ANFIS

The Adaptive Neuro-Fuzzy Inference System (ANFIS) is a hybrid intelligent system that combines the learning capabilities of artificial neural networks (ANN) with the intuitive knowledge representation of fuzzy inference systems (FISs) [32]. First proposed by Jang [22], the ANFIS demonstrates outstanding performance in modeling and predicting complex non-linear systems and is widely used in various engineering and scientific fields [33–36].

The ANFIS is typically based on a first-order Sugeno fuzzy model with two inputs and one output, and it consists of a five-layer feed-forward network structure. Each layer contains nodes that perform a specific function, receiving an input signal, processing it, and passing it through to the next layer.

For instance, consider a system with input variables x and y and an output variable f. We can establish two fuzzy rules as follows:

$$\begin{aligned} &\text{Rule 1}: \; x \in A_1 \wedge y \in B_1 \;\Rightarrow\; f_1 = p_1 x + q_1 y + r_1 \\ &\text{Rule 2}: \; x \in A_2 \wedge y \in B_2 \;\Rightarrow\; f_2 = p_2 x + q_2 y + r_2 \end{aligned} \tag{13}$$

Here, A_i and B_i are linguistic variables representing the fuzzy sets in the premise part, while p_i , q_i , and r_i are parameters that form the linear function in the consequent part.

The ANFIS learns the relationship between inputs and outputs based on these rules.

Layer 1: Fuzzification Layer

Each node in the first layer takes an input variable and calculates its degree of membership in a specific fuzzy set. Each node in this layer is adaptive, meaning its parameters, known as premise parameters, are adjusted during the learning process.

The output of node i, denoted as $O_{1,i}$, is calculated as follows:

$$O_{1,i} = \mu_{A_i}(x) \tag{14}$$

Here, x is the input signal and μ_{A_i} is the membership function of the fuzzy set A_i.

Layer 2: Rule Layer

Each node in the second layer is a fixed node. It calculates the firing strength (w_i) of a rule by multiplying the membership degrees from the previous layer. This corresponds to the logical AND operation in fuzzy logic, typically using the product T-norm operator.

$$O_{2,i} = w_i = \mu_{A_i}(x) \times \mu_{B_i}(y),\ i = 1, 2 \tag{15}$$

The output of each node represents how much a specific rule is activated for the current input.

Layer 3: Normalization Layer

Each node in the third layer is also a fixed node. It calculates the normalized firing strength, $\overline{w_i}$, by dividing the firing strength of each rule by the sum of all rules' firing strengths.

$$O_{3,i} = \overline{w_i} = \frac{w_i}{\sum_{j=1}^{n} w_j} \tag{16}$$

The normalized firing strength represents the relative contribution of each rule to the final output.

Layer 4: Defuzzification Layer

Each node in the fourth layer is an adaptive node. It computes the output of each rule by multiplying the normalized firing strength $\overline{w_i}$ from the previous layer by the rule's consequent linear function f_i.

$$O_{4,i} = \overline{w_i} f_i = \overline{w_i}(p_i x + q_i y + r_i) \tag{17}$$

The parameters in this layer, $\{p_i\ ,\ q_i\ , r_i\}$, are known as consequent parameters and are learned during the forward pass using the Least Squares Estimation method.

Layer 5: Output Layer

The fifth layer consists of a single fixed node that calculates the final output of the ANFIS, f, by summing the outputs of all rules from the fourth layer.

$$O_{5,1} = f = \sum_{i=1}^{m} \overline{w_i} f_i = \frac{\sum_{i=1}^{m} w_i f_i}{\sum_{i=1}^{m} w_i} \tag{18}$$

The ANFIS employs a hybrid learning algorithm to efficiently train both premise and consequent parameters. This algorithm consists of two stages: a forward pass and a backward pass.

Forward Pass: Consequent Parameter Learning Using Least Squares Estimation

In the forward pass, input data flows through the network to compute the output of each node and, ultimately, the final output. During this pass, the premise parameters are held constant, while the consequent parameters $\{p_i\ ,\ q_i\ , r_i\}$ are optimized using the Least Squares Estimation (LSE) method.

The final output f can be expressed as a linear combination of the consequent parameters:

$$f = \overline{w_1}(p_1 x + q_1 y + r_1) + \overline{w_2}(p_2 x + q_2 y + r_2) + \cdots = \sum_{i=1}^{m} \overline{w_i}(p_i x + q_i y + r_i) \tag{19}$$

For N training data points, this can be represented in matrix form:

$$Y = X\Theta \tag{20}$$

where Y is the vector of actual output values $(N \times 1)$.

X is a matrix composed of normalized firing strengths and input variables $(N \times (M \times (d+1)))$, where M is the number of rules and d is the number of inputs. Θ is the vector of unknown consequent parameters $((M \times (d+1)) \times 1)$.

Therefore, the optimal solution for the consequent parameters, Θ^*, can be calculated at once using the pseudo-inverse:

$$\Theta^* = \left(X^T X\right)^{-1} X^T Y \tag{21}$$

Backward Pass: Premise Parameter Learning Using Gradient Descent

In the backward pass, the consequent parameters are held constant. The error between the actual output and the ANFIS model's output is propagated backward through the network using the Gradient Descent algorithm to adjust the premise parameters.

Defining the error E as the squared error:

$$E = \left(f_{\text{target}} - f_{\text{output}}\right)^2 \tag{22}$$

The update rule for a premise parameter α (e.g., c or σ from a Gaussian function) is as follows, where η is the learning rate:

$$\alpha(t+1) = \alpha(t) - \eta \frac{\partial \alpha}{\partial E} \tag{23}$$

The error gradient $\frac{\partial \alpha}{\partial E}$ is calculated using the chain rule:

$$\frac{\partial \alpha}{\partial E} = \frac{\partial f}{\partial E} \cdot \frac{\partial \overline{w_i}}{\partial f} \cdot \frac{\partial w_i}{\partial \overline{w_i}} \cdot \frac{\partial \mu_i}{\partial w_i} \cdot \frac{\partial \alpha}{\partial \mu_i} \tag{24}$$

By calculating each term, the premise parameters are updated. This process is repeated to minimize the model's error.

2.5. FIS–CDC Model

To develop the FIS–CDC model, the process was carried out in the following order: data collection, definition of the stepwise CRI, and training using the ANFIS.

a. Data collection

In this study, data on the distance between two ships, collision time, and ship speed was utilized. Among these, the inter-ship distance and collision time can be obtained through CDC calculations. Once the simulation begins, CDC is designed to evaluate the distance between the two ships and assess the possibility of collision at 0.1 s intervals, providing the collision time if a collision occurs. Therefore, the values for inter-ship distance and collision time obtained through CDC's calculations were utilized as data. For ship speeds, the predefined values mentioned in Section 2.1 were used in this study.

b. Definition of the stepwise CRI

According to the four-level FIS-NC framework, the Danger stage (CRI $\geq$ 0.66) is defined as "Both ships are required to take such action as will best aid avoidance of a collision", while the Collision stage (CRI = 1.00) is defined as "Both ships have a collision". Building on these definitions, we calibrated the stage-wise collision risk index (CRI) for the FIS–CDC model. The CDC method calculates the theoretical collision distance for each relative bearing between two ships and compares it with their actual separation, thereby identifying both the timing and severity of the contact. Because the CDC method can pinpoint the exact moment of physical contact, its operative risk spectrum corresponds

to the upper end of the FIS-NC scale, spanning from Danger to *Collision*. Analysis of AIS data for bulk ships and container ships yielded the following CDC thresholds: collision at CRI = 1.00 (remaining distance to the other ship = 0 m); almost-collision for CRI $\geq$ 0.88 (remaining distance $\leq$ 50 m); and low collision risk for 0.77 $\leq$ CRI < 0.88 (remaining distance $\leq$ 100 m). These boundaries preserve consistency with the FIS-NC scheme while leveraging the CDC's ability to differentiate fine-grained levels of collision risk.

c. System development

The development process of the FIS–CDC model by learning the collected data with the ANFIS proceeded as follows. The collected trajectory data was segregated into bulk-ship and container-ship subsets; then, each subset was independently trained with the ANFIS to construct its corresponding FIS–CDC model.

First, fuzzy rules were extracted from the input–output data set:

$$Rule_i : \text{if } x_{1\ is}\ v \wedge x_{2\ is}\ \theta \wedge x_{3\ is}\ t_r \wedge x_{4\ is}\ d_r \Longrightarrow \text{CRI} = f_i(x_1, x_2, x_3, x_4) \tag{25}$$

where x_1, x_2, x_3, and x_4 are the input variables speed (v), intersection heading angle (θ_{ot}), collision time (t_c), and remaining distance (d_r), respectively, and v, θ_{0t}, t_c, and d_r are the fuzzy sets of the input variables (i.e., antecedent variables). The CRI is the consequent variable.

Following the Sugeno fuzzy model framework, the consequent function $f_i(x_1, x_2, x_3, x_4)$ for each rule is represented as a first-order polynomial in the four input variables. Partitioning each antecedent variable v, θ_{0t}, t_c, and d_r into three triangular membership functions, collision (C), almost-collision (AC), and near-collision (NC), generates $3^4 = 81$ fuzzy rules, the full set of which is summarized in Table 2.

$$f_i(x_1, x_2, x_3, x_4) = k_{i0} + k_{i1}x_1 + k_{i2}x_2 + k_{i3}x_3 + k_{i4}x_4 \tag{26}$$

$$(x_1,\ x_2,\ x_3,\ x_4) = (v,\ \theta_{0t},\ t_c,\ d_r) \tag{27}$$

Table 2. Components of the fuzzy inference rules for the FIS–CDC model.

Rule$_i$	v	θ	t_r	d_r
$Rule_1$	C	AC	NC	C
$Rule_2$	C	NC	AC	C
$Rule_3$	AC	C	NC	C
⋮	⋮	⋮	⋮	⋮
$Rule_{79}$	NC	C	AC	C
$Rule_{80}$	C	AC	C	NC
$Rule_{81}$	AC	C	C	NC

Second, parameter optimization employed a hybrid learning algorithm that combines Least Squares Estimation in the forward pass with Gradient Descent error back-propagation in the backward pass [14]. During the forward pass, the premise parameters are held fixed while the consequent parameters are determined by the Least Squares method. During the backward pass, the output error is propagated to the input layer and the premise parameters are updated via Gradient Descent. The entire dataset was divided into 60% for training, 20% for validation, and 20% for testing. The validation subset was used to monitor generalization and implement early stopping before overfitting, whereas the test subset provided a fully independent assessment of model performance. Layer-wise operations and the corresponding learning equations are detailed in Section 2.5. Once parameter optimization was completed, the learning procedure yielded the final Sugeno-type FIS–CDC architecture depicted in Figures 5–7.

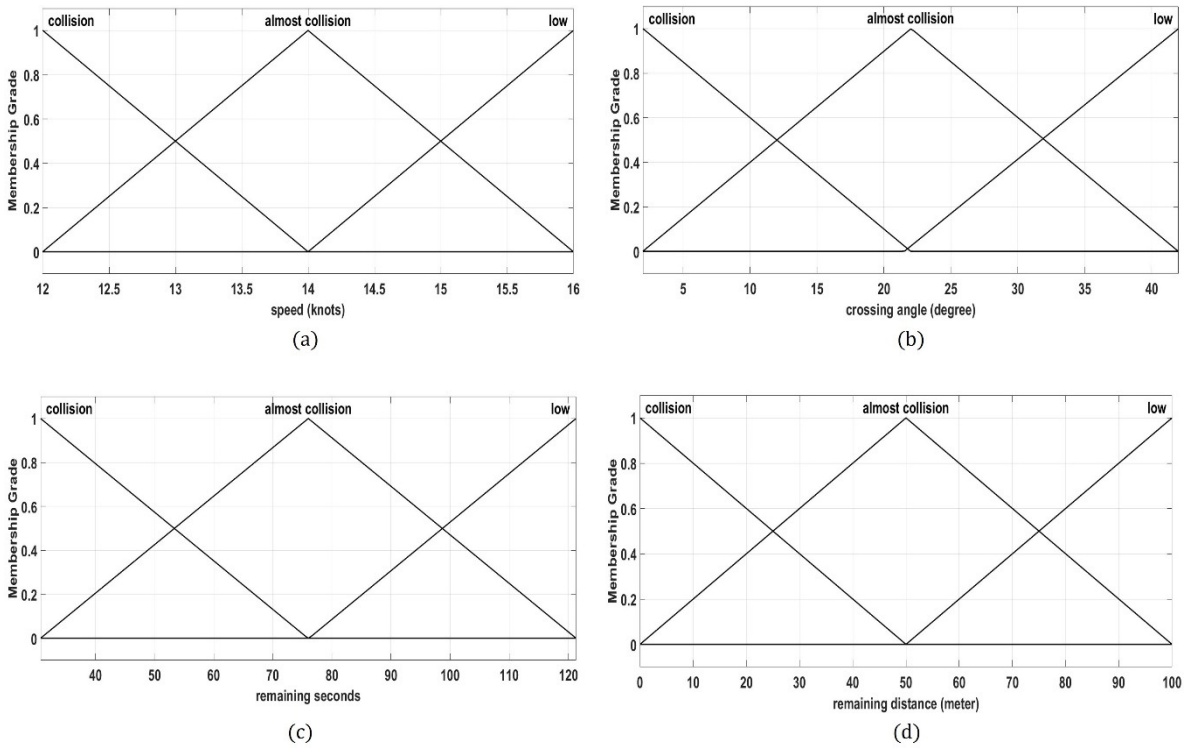


Figure 5. Triangular fuzzy membership functions of the FIS–CDC model for the bulk ship. (**a**) Speed, (**b**) crossing angle, (**c**) remaining seconds, and (**d**) remaining distance.

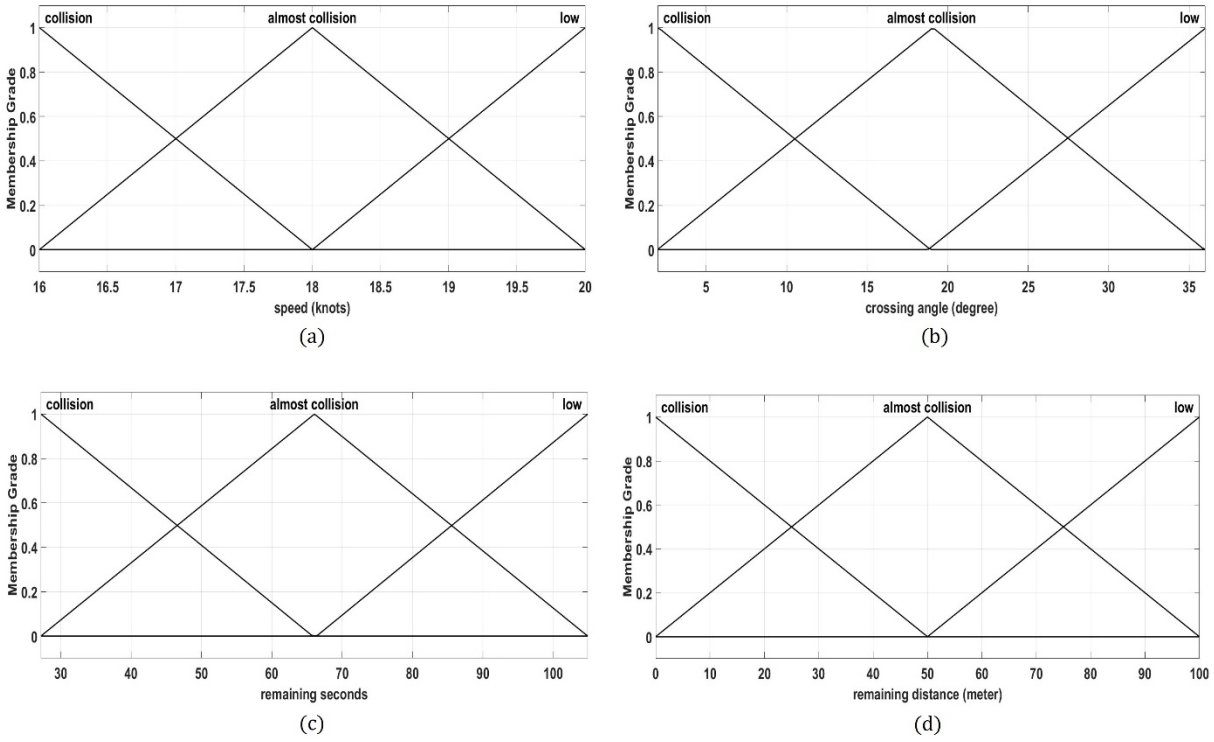


Figure 6. Triangular fuzzy membership functions of the FIS–CDC model for the container ship. (**a**) Speed, (**b**) crossing angle, (**c**) remaining seconds, and (**d**) remaining distance.

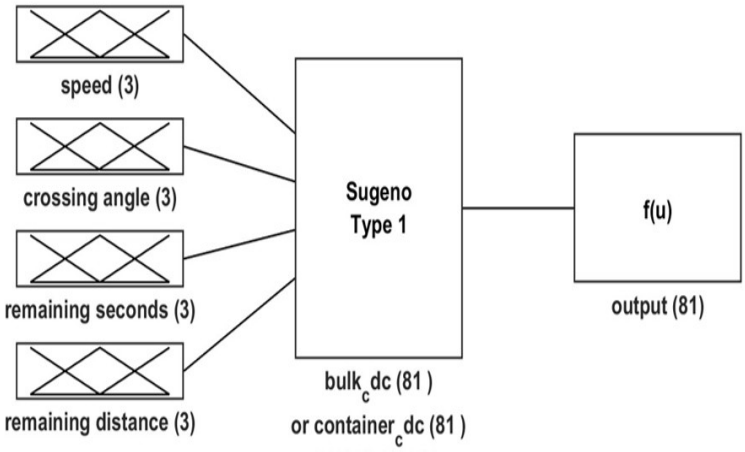


Figure 7. Structure of the FIS–CDC model.

Using the bulk-ship dataset, ANFIS training produced a collision-risk inference system characterized by triangular membership functions. The resulting membership surfaces and rule partitions are depicted in Figure 5. Applying the same training procedure to the container-ship dataset generated an analogous model, the details of which are presented in Figure 6.

Figure 7 depicts the Sugeno-type fuzzy inference system (FIS–CDC) developed in this study. The antecedent layer comprises the four kinematic variables speed, crossing angle, remaining seconds, and remaining distance. Each variable is fuzzified into three triangular membership functions representing the linguistic terms collision, almost-collision, and low, yielding a total of 81 fuzzy rules through full factorial combination.

During inference, the firing strength of each rule is obtained by multiplying the four membership grades associated with the current input. The crisp collision risk index (CRI) is then produced by aggregating these rule outputs with their firing strengths as weights. In other words, the crisp output of the proposed FIS–CDC model is derived by fully aggregating the firing strengths of all 81 rules produced from the three-level partitions of v, θ, t_r, and d_r. The resulting weighted-average (WA) defuzzification is expressed by the function $\hat{f}$ in Equation (28).

$$\hat{f} = \sum_{i=1}^{81} \frac{\mu(k_{i,1})\, k_{i,1} + \mu(k_{i,2})\, k_{i,2} + \mu(k_{i,3})\, k_{i,3} + \mu(k_{i,4})\, k_{i,4}}{\mu(k_{i,1}) + \mu(k_{i,2}) + \mu(k_{i,3}) + \mu(k_{i,4})} \tag{28}$$

By combining local linear consequents with WA defuzzification, the system obtains a smooth and computationally efficient CRI surface, making it well-suited for real-time collision-avoidance support aboard bulk ships and container ships.

3. Results

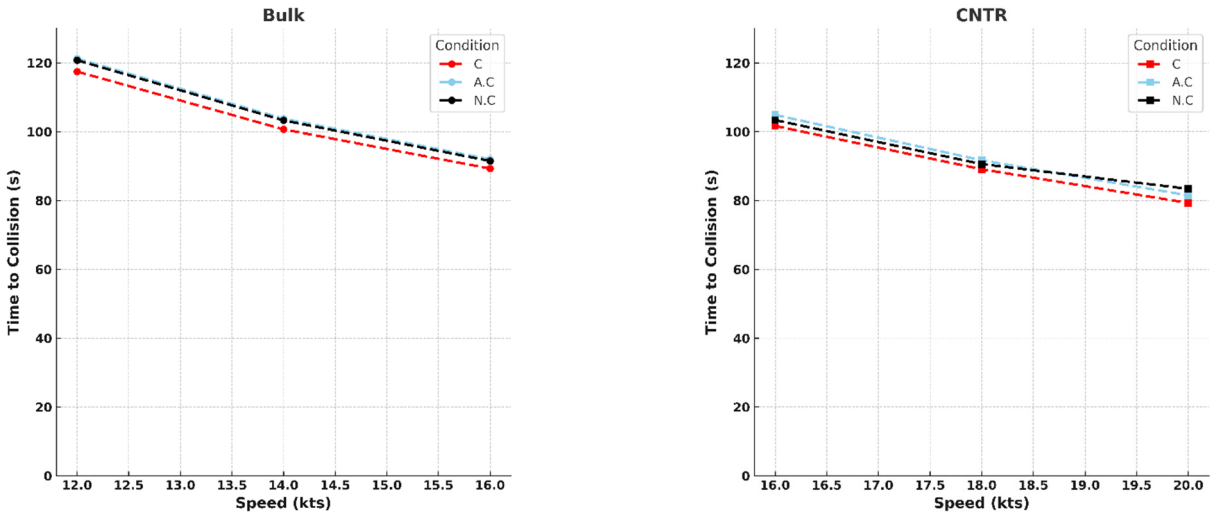
3.1. Results of CDC

This study aimed to systematize the fusion of CDC and the ANFIS. To achieve this, we first conducted a collision assessment using CDC for bulk ships and container ships. The collision assessment results are presented in Table 3 and Figure 8.

In Table 3, C denotes a collision, AC signifies an almost-collision where the passing distance between two ships is within 0.2 L, and NC indicates a near-collision where two ships passed at a distance of 0.4 L. Figure 8 shows the collision time in the maximum θ_{ot} in each case (C, AC, and NC). The horizontal and vertical axes are speed and collision time, respectively.

Table 3. Results of the collision assessment.

Type	Speed	Result	θ_{ot} (°)	Time (s)
Bulk	12 kts	C	28	117.5
		A.C	36	121.3
		N.C	42	120.8
Bulk	14 kts	C	28	100.7
		A.C	36	103.8
		N.C	42	103.3
Bulk	16 kts	C	28	89.3
		A.C	36	92.0
		N.C	42	91.5
CNTR	16 kts	C	24	101.7
		A.C	30	104.9
		N.C	34	103.4
CNTR	18 kts	C	24	89.1
		A.C	30	91.7
		N.C	34	90.6
CNTR	20 kts	C	24	79.3
		A.C	30	81.5
		N.C	36	83.3



(**a**) (**b**)

Figure 8. Time at the maximum θ_{ot} for each case. (**a**) Bulk ship, (**b**) container ship.

As shown in Figure 8 and Table 3, t_c decreases as the ship's speed increases. Additionally, it can be observed that t_c for container ships is shorter than that for bulk ships.

A longer t_c indicates that more time is available for collision avoidance, while a shorter t_c signifies a more urgent situation and a higher risk of collision. Therefore, we can see that the risk of collision increases as the ship's speed increases. For example, under 'C' conditions, t_c for bulk ships drops from 117.5 s at 12 kts to 89.3 s at 16 kts. Similarly, for container ships under 'C' conditions, it decreases from 101.7 s at 16 kts to 79.3 s at 20 kts.

Generally, the C situation exhibited the shortest t_c, while, interestingly, the AC case yielded the longest t_c in some scenarios, possibly due to the proximity criteria defined for AC situations. The reason the collision occurrence time in NC cases did not increase compared to AC cases is likely due to the definition of NC, where two ships pass within 100 m, even if the intersection angle between the two ships is large enough to otherwise delay the collision point. This factor seems to have played a role in this result.

Additionally, bulk ships exhibited a slightly wider range of t_c compared with container ships. This indicates that, when two ships are in close proximity, it is more difficult for bulk ships to avoid a collision. Next, a more in-depth analysis was conducted using detailed data for each ship type.

3.2. Result of Collision Assessment of Bulk Ships

For bulk ships, the collision assessment analyzed t_c across various θ_{ot} under three speed conditions: 12 kts, 14 kts, and 16 kts. The analysis results are presented in Table 4 and Figure 9.

Table 4. Result of collision assessment of bulk ships.

$\theta_{ot}(^\circ)$	12 kts		14 kts		16 kts	
	Result	**Time**	**Result**	**Time**	**Result**	**Time**
2	C	38.9	C	34	C	30.7
4	C	54.2	C	47.2	C	42.2
6	C	64.4	C	55.8	C	49.9
8	C	72.4	C	62.6	C	55.8
10	C	79.1	C	68.3	C	60.8
12	C	85.1	C	73.4	C	65.2
14	C	90.4	C	77.9	C	69.2
16	C	93.9	C	81	C	71.9
18	C	96	C	82.8	C	73.4
20	C	99.6	C	85.6	C	76.0
22	C	103.9	C	89.3	C	79.2
24	C	108.3	C	93	C	82.5
26	C	112.8	C	96.7	C	85.8
28	C	117.5	C	100.7	C	89.3
30	AC	109.8	AC	94.2	AC	83.5
32	AC	113.5	AC	97.3	AC	86.3
34	AC	117.3	AC	100.5	AC	89.1
36	AC	121.3	AC	103.8	AC	92.0
38	NC	114.2	NC	97.8	NC	86.7
40	NC	117.5	NC	100.6	NC	89.1
42	NC	120.8	NC	103.3	NC	91.5

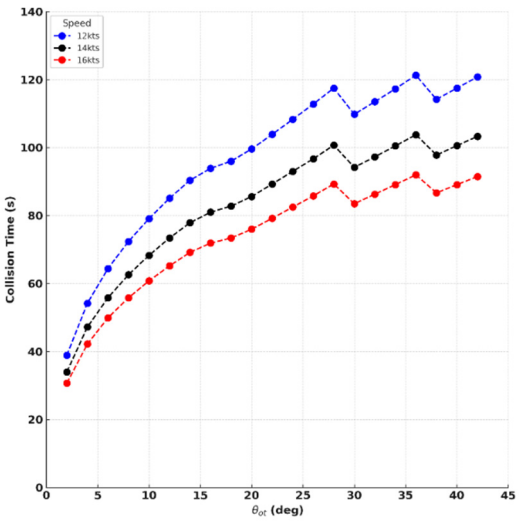


Figure 9. t_c, t_{ac}, and t_{nc} based on θ_{ot} and conditions in bulk ships.

In Figure 9, the horizontal and vertical axes denote θ_{ot} and t_c, t_{ac}, and t_{nc}, respectively, and blue, black, and red marks and lines indicate ship speeds.

First, analyzing t_c, t_{ac}, and t_{nc} trends for bulk ships reveals a consistent pattern. As the speed increases, collision times decrease across all θ and collision conditions. For example, at θ_{ot} of 20°, t_c dropped significantly from 99.6 s at 12 kts to 76 s at 16 kts. This indicates that higher bulk ship speeds reduce the time available before a collision, consequently increasing the collision risk in similar scenarios. This outcome is intuitively sound and shows that CDC predicts shorter estimated t_c values at higher speeds, reflecting greater risk.

Second, within each speed condition, there was a general tendency for t_c to increase as θ_{ot} increased. This indicates that a larger θ_{ot} provides more time for collision avoidance. For example, at 12 kts, 38.9 s remained at θ_{ot} of 2°, but this increased significantly to 117.5 s at θ_{ot} of 28°. This trend was consistently observed across all speed ranges. However, for NC cases, t_{nc} appeared somewhat shorter compared with t_{ac}. This was subsequently analyzed in detail as follows.

Within the C range, defined by θ_{ot} from 2° to 28°, a clear linear increase in t_c was observed as θ_{ot} widened. For instance, at 14 kts, θ_{ot} of 2° leaves 34 s until collision, while θ_{ot} of 28° provides 100.7 s. This demonstrates that the collision assessment model accurately predicts that a greater θ_{ot} directly affords more time for evasive maneuvers. In the AC range, from θ_{ot} of 30° to 36°, the trend of increasing t_{ac} with a larger θ_{ot} persists. The NC range, covering θ_{ot} from 38° to 42°, was observed to be the narrowest compared with the other cases. Likewise, it was confirmed that t_{nc} also increased as θ_{ot} increased. The reason that the t_{nc} of the maximum θ_{ot} in NC cases is smaller than in AC cases stems from NC's classification criteria. NC includes a standard in which a situation is deemed low risk if the passing distance between two ships is within 0.4 L. This distance criterion overrides the actual θ_{ot}, leading to the maximum θ_{ot} for NC cases being limited despite a physically larger θ_{ot} being possible.

Lastly, for bulk ships, even with an increase in speed, there was no change in the maximum θ_{ot} for the C, AC, and NC conditions (only the time was reduced). This result is somewhat different from what was observed with container ships.

3.3. Result of Collision Assessment of Container Ships

For container ships, the collision assessment results were analyzed at three speed conditions (16 kts, 18 kts, and 20 kts), as presented in Table 5 and Figure 10.

As shown in Figure 10, the horizontal and vertical axes denote θ_{ot} and t_c, t_{ac}, and t_{nc}, respectively, and blue, black, and red marks and lines indicate ship speeds.

First, similar to bulk ships, container ships also showed a consistent trend. As the speed increased, t_c decreased across all θ_{ot} and C conditions. For instance, in C conditions, t_c dropped from 101.7 s at 16 kts to 79.3 s at 20 kts. This means that, for container ships as well, higher speeds lead to less t_c, indicating an increased risk.

Second, consistent with bulk ships, within each speed condition, there was a general tendency for t_c, t_{ac}, and t_{nc} to increase as θ_{ot} increased. This indicates that the collision assessment model predicts that larger θ_{ot} values provide more time for collision avoidance in container ship scenarios. For example, at 16 kts, θ_{ot} of 2° showed 33.4 s remaining, while θ_{ot} of 24° showed 101.7 s.

Third, a detailed analysis was conducted on the change in t_c according to various collision conditions. In the C range, specifically for θ_{ot} from 2° to 24°, a clear linear increase in t_c was observed as θ_{ot} increased. Moving to the AC range, spanning θ_{ot} from 26° to 30°, the increase in t_{ac} with θ_{ot} persists. However, the rate of this increase is somewhat gentler compared with the C range. Finally, in the NC range, covering θ_{ot} from 32° to 36°, a phenomenon similar to that observed with bulk ships was noted: t_{nc} either decreased or plateaued. For instance, at 16 kts, the time was 104.9 s at 30° and 99.9 s at 32°, illustrating

this trend. The overall θ_{ot} range was narrower compared with bulk ships, but, uniquely, the maximum θ_{ot} value for NC cases changed when the speed was 20 kts.

Table 5. Result of collision assessment of container ships.

Angle	16 kts		18 kts		20 kts	
	Result	Time	Result	Time	Result	Time
2	C	33.4	C	29.9	C	27.2
4	C	46.6	C	41.4	C	37.4
6	C	55.8	C	49.5	C	44.5
8	C	63.2	C	56	C	50.2
10	C	69.7	C	61.6	C	55.2
12	C	75.4	C	66.6	C	59.6
14	C	78.9	C	69.9	C	62.8
16	C	82.2	C	72.3	C	64.6
18	C	87	C	76.5	C	68.3
20	C	91.7	C	80.6	C	71.9
22	C	96.5	C	84.7	C	75.5
24	C	101.7	C	89.1	C	79.3
26	AC	96.4	AC	84.6	AC	75.3
28	AC	100.4	AC	88.1	AC	78.4
30	AC	104.9	AC	91.7	AC	81.5
32	NC	99.9	NC	87.6	NC	77.9
34	NC	103.4	NC	90.6	NC	80.6
36					NC	83.3

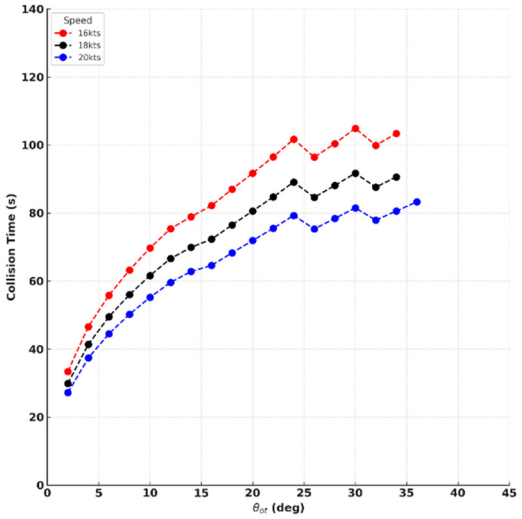


Figure 10. t_c, t_{ac}, and t_{nc} based on θ_{ot} and conditions in container ships.

3.4. Comparative Analysis of Bulk Ships and Container Ships

In this section, we analyze the differences between the two types of ships. Figure 11 illustrates the collision time according to the ship speed and θ_{ot} for each ship.

By comparatively analyzing the collision assessment model results for bulk ships and container ships, we can understand the characteristics of collision assessment for each ship type.

Comparing the t_c for both ship types at the same speed of 16 kts, it was generally observed that container ships had a longer t_c than bulk ships. For example, at 16 kts and θ_{ot} of 24°, the t_c of bulk ships was 82.5 s while the t_c of container ships in the same θ_{ot} was 101.7 s. This suggests that, under the same speed conditions, container ships have more

time for collision avoidance compared with bulk ships. Therefore, the following analysis was conducted to examine how t_c changes as the ship speed increases for both ship types.

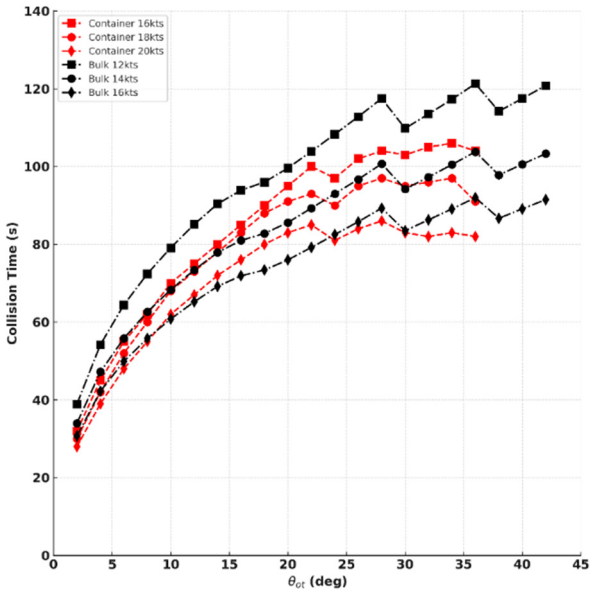
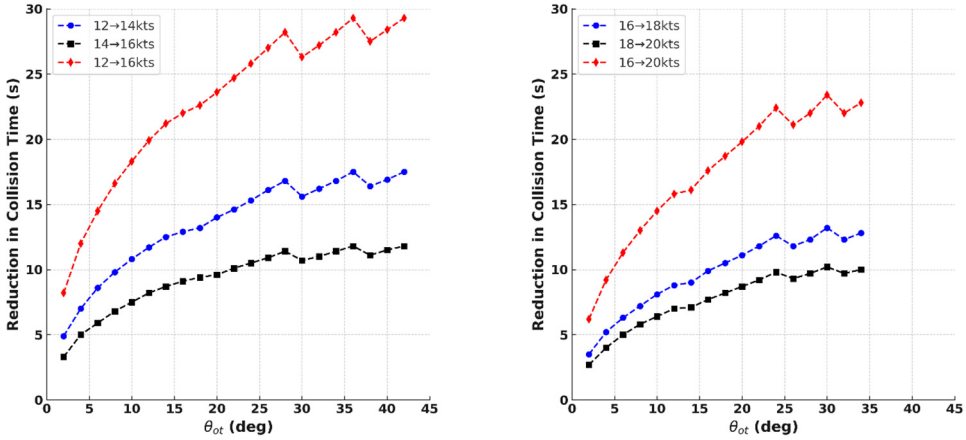


Figure 11. Comparison of results based on the speed between two ships.

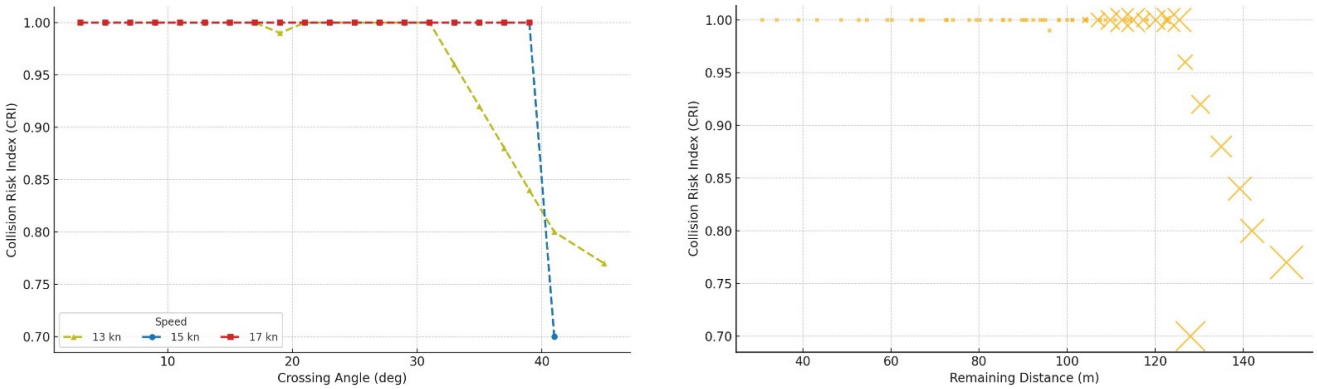
In Figure 12, the horizontal and vertical axes represent θ_{ot} and the decrease in t_c, respectively. The blue and black lines indicate the time reduction as the speed increases by 2 knots, while the red line shows the total time reduction between the lowest and highest speeds. As shown in Figure 13, the bulk ship exhibits a larger reduction in t_c. For the bulk ship, t_c decreased by up to approximately 30 s as the speed increased. In contrast, the container ship showed a maximum reduction of about 23 s. The average reduction in time to collision for both ship types is summarized in Table 6.

As shown in Table 6, when the ship speed increased by 4 knots, the average time reduction rate of the result value for the bulk ship was approximately 23.13%, which is about 1.86% higher than that of the container ship (21.27%). Although this difference is not particularly significant, it is noteworthy that the bulk ship exhibited a higher Δt_c despite being shorter in length. This can be attributed to its lower speed and inferior maneuverability compared with the container ship.



(**a**) (**b**)

Figure 12. Comparison of time reduction rates. (**a**) Bulk ships, (**b**) container ships.



(a) (b)

Figure 13. The key behavioral patterns of the FIS–CDC model for bulk ships. (**a**) CRI vs. crossing angle at different speeds, (**b**) CRI vs. remaining distance.

Table 6. The average reduction in time.

Ship Type	Initial Speed	Final Speed	ΔSpeed	Av. Rate of Reduction Time
Bulk	12 kts	16 kts	4 kts	23.13%
Container	16 kts	20 kts	4 kts	21.27%

Both ship types exhibited a common trend where t_c, t_{ac}, and t_{nc} decreased as the speed increased and also a common trend where t_c, t_{ac}, and t_{nc} increased as θ_{ot} increased. Specifically, under the same speed conditions, container ships generally secured a longer collision time compared with bulk ships, suggesting more available time for avoidance. Conversely, in terms of collision time reduction due to increased speed, bulk ships exhibited a slightly higher average reduction rate and a broader maximum reduction range than container ships, indicating a greater sensitivity to speed changes regarding the available collision avoidance time.

3.5. Simulation-Based Robustness Evaluation of the FIS–CDC Model

To evaluate the robustness of the developed FIS–CDC model under a previously unobserved data distribution, all even-numbered speeds and crossing angles used during the training phase were excluded and the parameters were redefined using only odd-numbered intervals. Two distinct ship categories—bulk and container ships—were considered. Based on all combinations of these parameters, a comprehensive set of test scenarios was generated. Initial positions and relative courses were configured according to the procedure detailed in Section 2.1. Subsequently, the ship-motion equations described in Section 2.2 were numerically integrated to obtain time-series trajectories. The CDC algorithm outlined in Section 2.3 was then applied to extract the remaining time and remaining distance to the predicted point of collision. These variables—namely speed, θ_{ot}, remaining time, and remaining distance—were collectively provided as inputs to the FIS–CDC model, which inferred the corresponding CRI.

To further assess the performance of the FIS–CDC model in the context of bulk ships, a dedicated simulation campaign was conducted using the parameter combinations introduced in Section 2. The key behavioral patterns are visualized in Figure 13.

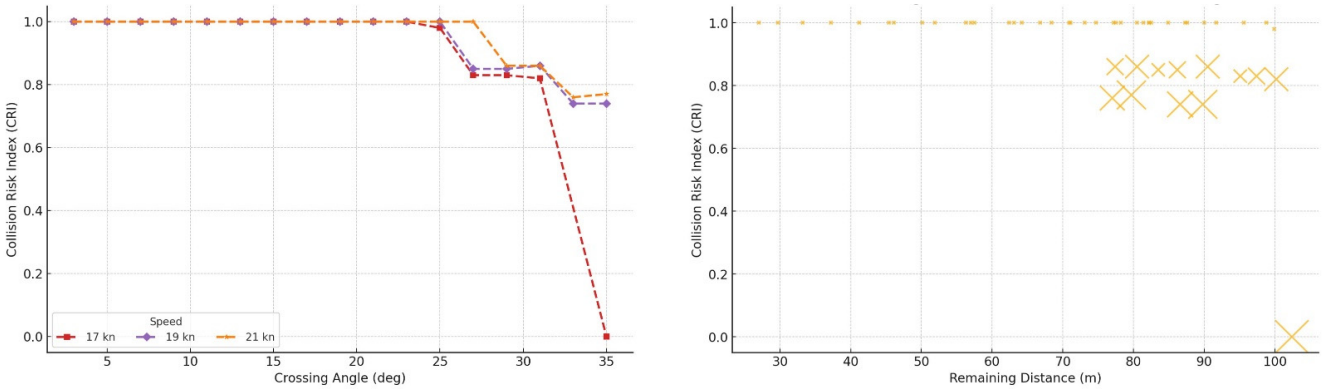
Figure 13a depicts the collision-risk surface across varying crossing angles at three representative service speeds: 13, 15, and 17 kts. At 17 kts, the CRI remains constant at approximately 1.00 over the entire angular spectrum, indicating that a collision is consistently predicted regardless of encounter geometry. In contrast, the 15 kt trajectory reaches a CRI of 1.00 only when the crossing angle exceeds approximately 40°, suggesting that an

elevated collision risk manifests primarily during wide-angle encounters. At the lowest speed, 13 kts, the risk profile exhibits a gradual increase beyond 30°, with CRI values rising from 0.99 to 1.00. This trend implies that, at lower speeds, the likelihood of collision intensifies more rapidly as the encounter angle widens.

Figure 13b complements this analysis by plotting CRI as a function of remaining distance, with marker sizes scaled to the remaining time to the potential point of collision. Scenarios with an initial separation of less than 100 m almost invariably yield CRI values close to 1.00, signifying imminent collision. Conversely, cases initiated at distances exceeding 120 m—accompanied by longer temporal buffers—tend to exhibit elevated but sub-maximal CRI values ranging between 0.70 and 0.90.

To evaluate the performance of the proposed FIS–CDC model for container ships, a separate set of simulations was conducted following the same procedure, with the ship type set to container ships. Figure 13 presents the key outcomes derived from the consolidated raw data.

Figure 14a presents the collision-risk surface across varying crossing angles at three representative speeds: 17, 19, and 21 kts. At all three speeds, the CRI remains fixed at 1.00 for crossing angles below 25°, indicating that, in narrow-angle encounters, collisions are virtually unavoidable due to the lack of a maneuvering margin. As the speed increases (21 kts > 19 kts > 17 kts), the critical crossing angle beyond which the CRI begins to drop rises slightly. However, even at wider angles, high-speed navigation does not substantially reduce the collision risk. In contrast, at 17 kts, the CRI approaches zero when the crossing angle reaches approximately 35°, confirming that low-speed, wide-angle encounters are the most effective for collision avoidance.



(a) (b)

Figure 14. The key behavioral patterns of the FIS–CDC model for container ships. (**a**) CRI vs. crossing angle at different speeds, (**b**) CRI vs. remaining distance.

Figure 14b complements this analysis by visualizing the relationship between remaining distance and CRI, with marker sizes proportional to the remaining time until the potential point of collision. Within the 25–70 m range, most cases are concentrated at CRI = 1.00, suggesting that collisions are imminent when spatial margins are insufficient. In the 80–100 m range, the risk levels moderate to CRI values between 0.70 and 0.90; however, in scenarios involving large crossing angles, the likelihood of collision remains high. Notably, a single case near 100 m with the CRI falling to zero demonstrates that sufficient angular separation can fully eliminate the collision risk.

4. Discussion

The simulation results presented in the previous section demonstrate that the proposed FIS–CDC model effectively captures the non-linear risk dynamics across diverse ship types

and encounter conditions. Based on these findings, this section discusses the reliability, interpretability, and operational applicability of the newly defined CRI thresholds and the four-variable CDC inference model.

The CRI thresholds proposed in this study were found to correspond naturally with the empirical distribution of remaining distance and remaining time. When the remaining distance reaches 0 m, the CRI converges exactly to 1.00, clearly indicating a collision state. Within a 0.2 L range, most cases exhibit CRI values exceeding 0.88, which is consistent with almost-collision (AC) levels frequently observed in high-risk interactions. In the range of 0.2 L to 0.4 L, the CRI values gradually decline to between 0.77 and 0.88, denoting a near-collision (NC) buffer zone.

This intuitive mapping between spatial–temporal thresholds and risk levels offers a potential foundation for timely interpretation by navigators and VTS operators. It is expected to aid in more informed evasive decision-making. However, we acknowledge that the precise interpretation and operational implementation of such concepts (e.g., 'timely' action, 'appropriate' decision) remain inherently context-dependent, varying significantly based on factors like vessel characteristics, environmental conditions, and specific encounter geometries.

Building upon this reliable CRI scheme, the CDC employs a four-variable input structure—speed, crossing angle, remaining distance, and remaining time—to infer collision risk. In stark contrast to traditional indicators such as Distance to Closest Point of Approach (DCPA) and Time to Closest Point of Approach (TCPA), which often rely solely on static distance and time metrics calculated from GPS antenna positions, our inclusion of ship speed and crossing angle enables the CDC to reflect dynamic, ship-specific kinematic characteristics more accurately. This fundamentally enhances the model's ability to capture the real-time, evolving nature of collision scenarios. As a result, the system can identify high-risk scenarios at earlier stages. For example, in a situation where a container ship is sailing at 21 kts with a 20° crossing angle and has 60 m and 15 s remaining to a potential collision, conventional indicators might suggest a low-risk encounter. However, the CDC immediately yields a CRI exceeding 0.95, issuing an early warning by integrating all four critical variables. This multidimensional risk representation enhances the system's ability to distinguish between different ship types—such as high-maneuverability container ships and low-maneuverability bulk ships—and to tailor collision-avoidance recommendations accordingly.

Moreover, the four-variable CDC architecture effectively absorbs ship-specific maneuvering traits during the training process, enabling straightforward extension to various ship classes—including bulk carriers, container ships, and tankers—without requiring structural changes to the model. Owing to its compact input space, the system maintains a low computational burden, making it well-suited for time-sensitive applications such as collision warnings and real-time autonomous navigation.

Furthermore, the FIS–CDC system can be particularly valuable in two practical operational contexts. First, within the MASS environment, the model can be embedded into onboard navigation controllers to provide real-time collision risk inference based on vessel-specific characteristics. For instance, during autonomous transit through high-traffic coastal waters, the system can deliver early collision warnings even in close-encounter scenarios involving diverse ship types. Second, in VTS operations, the CDC-based risk index can assist shore-based operators in simultaneously monitoring multiple vessels, particularly under limited visibility conditions or communication delays. For example, a VTS center managing operations in a major port such as Busan could leverage CRI outputs to prioritize guidance or advisory messages for vessels with elevated collision risk based on CDC-inferred interaction trajectories. These practical examples illustrate

how the model supports timely and risk-informed decision-making in both ship-side and shore-side domains.

In conclusion, the CRI thresholds and four-variable CDC framework introduced in this study provide a robust and interpretable foundation for rapid and accurate collision risk assessment. The demonstrated ability to integrate multiple risk factors and ship dynamics underscores the framework's potential as a core component in next-generation autonomous and semi-autonomous navigation systems aimed at ensuring maritime safety in increasingly congested sea environments.

Despite these significant contributions, this study has several limitations that should be addressed in future research. Although research was conducted on bulk carriers and container ships, directly applying these results to other ship types remains limited. The most significant limitation is the substantial influence of OS and TS sizes on collision assessment results. The current simulations were conducted with both ships having identical specifications or were limited to specific models for consistency. However, the real maritime environment consists of ships with diverse sizes and characteristics, and changes in ship specifications directly impact collision assessment outcomes. Furthermore, this study does not account for external environmental factors such as weather and sea conditions, which can significantly influence ship dynamics and decision-making. To address this issue, future research should continue to accommodate a wider range of ship sizes.

Furthermore, for application to various individual ships, research on CDC algorithms must be conducted to enable real-time computation. Additionally, continuous research is required targeting a wider range of opposing ships under diverse scenarios. Addressing these limitations, including the integration of environmental factors and the explicit consideration of varying cargo conditions and their impact on ship maneuverability, will be crucial for developing more robust and widely applicable real-time collision avoidance systems in the evolving field of MASSs.

5. Conclusions

This study proposed an enhanced regional collision risk prediction system by integrating the CDC method with the ANFIS, aiming to support timely and accurate decision-making in vessel traffic services and MASSs. The model was constructed and validated through detailed simulations involving bulk and container ships under various crossing angles and speeds.

The simulation results reveal that the time to collision (t_c) consistently decreases as the ship speed increases and increases with a wider θ_{ot} collision range. Furthermore, bulk carriers showed a wider θ_{ot} collision range and greater sensitivity to speed changes compared with container ships, indicating higher collision risk in close-proximity scenarios. The proposed FIS–CDC system effectively captured these dynamics and demonstrated robustness in differentiating risk levels, even under untrained parameter conditions.

By employing four key input variables—ship speed, crossing angle, remaining distance, and remaining time—the FIS–CDC system outperformed traditional CPA-based metrics in reflecting ship-specific maneuvering characteristics and predicting imminent collisions. Its compact structure also enables real-time application with low computational cost, making it suitable for operational deployment.

Nevertheless, the scope of the present study is limited to two ship types of fixed size. Future research should address a broader spectrum of ship classes, sizes, and sea conditions to improve generalizability. Real-time CDC algorithms and extensive validation with AIS data will also be essential to realize a fully autonomous, reliable risk prediction system for diverse maritime environments.

Author Contributions: Conceptualization, H.N. and H.-J.L.; methodology, H.N. and H.-J.L.; software, H.N. and H.-J.L.; validation, H.N. and H.-J.L.; formal analysis, H.-J.L.; investigation, H.N. and H.-J.L.; resources, H.N. and H.-J.L.; data curation, H.-J.L.; writing—original draft preparation, H.N. and H.-J.L.; writing—review and editing, H.-J.L.; visualization, H.N. and H.-J.L.; supervision H.-J.L.; project administration H.-J.L. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding author.

Conflicts of Interest: The authors declare no conflicts of interest.

Nomenclature

Symbols	Definitions
CDC	Computed distance at collision
OS	Own ship
TS	Target ship
VOS	Virtual OS
δ_{OS}	OS rudder angle
t_s	Simulation time
θ_{ot}	Crossing angle between the OS and TS
L	Ship length
AD	Actual distance between the OS and TS
x_{os}, y_{os}	Coordinates of the OS in an Earth-fixed coordinate system
x_{os}^f, y_{os}^f	Coordinates of the OS forecastle point in the Earth-fixed coordinate system
x_{vos}, y_{vos}	Coordinates of the VOS in the Earth-fixed coordinate system
x_{vos}^f, y_{vos}^f	Coordinates of the VOS forecastle point in the Earth-fixed coordinate system
x_{ts}, y_{ts}	Coordinates of the TS in the Earth-fixed coordinate system
ψ_{os}, ψ_{ts}	Headings of the OS and TS
x_c, y_c	Coordinates of the collision point in the Earth-fixed coordinate system
t_c, t_{ac}, t_{nc}	Time of collision, almost-collision, and near-collision

References

1. Zhang, L.; Wang, H.; Meng, Q.; Xie, H. Ship accident consequences and contributing factors analyses using ship accident investigation reports. *Proc. Inst. Mech. Eng. Part O J. Risk Reliab.* **2019**, *233*, 35–47. [CrossRef]
2. Korean Maritime Safety Tribunal. Status of Marine Accidents. Available online: https://www.kmst.go.kr/eng/page.do?menuIdx=227 (accessed on 6 May 2025).
3. Lee, H.J.; Furukawa, Y. Timing of stand-on ship actions for cooperative collision avoidance under calm water condition. *Ocean Eng.* **2022**, *263*, 112340. [CrossRef]
4. Lee, H.J.; Furukawa, Y.; Park, D.J. Seafarers' awareness-based domain modelling in restricted areas. *J. Navig.* **2021**, *74*, 1172–1188. [CrossRef]
5. Lee, H.J.; Park, D.J. SASD Modeling Using an ANFIS to Prevent the Collision of MASS in Restricted Areas. *J. Mar. Sci. Eng.* **2022**, *10*, 961. [CrossRef]
6. Zhou, Y.; Du, W.; Liu, J.; Li, H.; Grifoll, M.; Song, W.; Zheng, P. Determination of ship collision avoidance timing using machine learning method. *Sustainability* **2024**, *16*, 4626. [CrossRef]
7. Zheng, M.; Zhang, K.; Han, B.; Lin, B.; Zhou, H.; Ding, S.; Yang, Y. An improved VO method for collision avoidance of ships in open sea. *J. Mar. Sci. Eng.* **2024**, *12*, 402. [CrossRef]
8. Lyu, H.; Ma, X.; Tan, G.; Yin, Y.; Sun, X.; Zhang, L.; Song, J. Identification of complex multi-vessel encounter scenarios and collision avoidance decision modeling for MASSs. *J. Mar. Sci. Eng.* **2024**, *12*, 1289. [CrossRef]
9. Shen, Y.; Liao, Z.; Chen, D. Differential Evolution Deep Reinforcement Learning Algorithm for Dynamic Multiship Collision Avoidance with COLREGs Compliance. *J. Mar. Sci. Eng.* **2025**, *13*, 596. [CrossRef]
10. Gao, M.; Shi, G.Y. Ship collision avoidance anthropomorphic decision-making for structured learning based on AIS with Seq-CGAN. *Ocean Eng.* **2020**, *217*, 107922. [CrossRef]

11. Chun, D.H.; Roh, M.I.; Lee, H.W.; Ha, J.; Yu, D. Deep reinforcement learning based collision avoidance for an autonomous ship. *Ocean Eng.* **2021**, *234*, 109216. [CrossRef]
12. Porathe, T. Maritime Autonomous Surface Ships (MASS) and the COLREGS: Do weneed quantified rules or is "the ordinary practice of seamen" specific enough? *TransNav Int. J. Mar. Navig. Saf. Sea Transp.* **2019**, *13*, 511–518. [CrossRef]
13. Huang, Y.; Chen, L.; Chen, P.; Negenborn, R.R.; Van Gelder, P.H.A.J.M. Ship collision avoidance methods: State-of-the-art. *Saf. Sci.* **2020**, *121*, 451–473. [CrossRef]
14. Namgung, H.; Kim, J.S. Collision risk inference system for maritime autonomous surface ships using COLREGs rules compliant collision avoidance. *IEEE Access* **2021**, *9*, 7823–7835. [CrossRef]
15. Seo, C.; Noh, Y.; Abebe, M.; Kang, Y.J.; Park, S.; Kwon, C. Ship collision avoidance route planning using CRI-based A∗ algorithm. *Int. J. Nav. Archit. Ocean. Eng.* **2023**, *15*, 100551. [CrossRef]
16. Hu, Y.; Zhang, A.; Tian, W.; Zhang, J.; Hou, Z. Multi-ship collision avoidance decision-making based on collision risk index. *J. Mar. Sci. Eng.* **2020**, *8*, 640. [CrossRef]
17. Hasegawa, K.; Kouzuki, A.; Muramatsu, T.; Komine, H.; Watabe, Y. Ship Auto-Navigation Fuzzy Expert System (SAFES). *J. Soc. Nav. Archit. Jpn.* **1989**, *166*, 445–452. [CrossRef] [PubMed]
18. Lee, H.; Rhee, K. Development of Collision Avoidance System by Using Expert System and Search Algorithm. *Int. Shipbuild. Prog.* **2001**, *48*, 197–212.
19. Ahn, J.-H.; Rhee, K.-P.; You, Y.-J. A Study on the Collision Avoidance of a Ship Using Neural Networks and Fuzzy Logic. *Appl. Ocean Res.* **2012**, *37*, 162–173. [CrossRef]
20. Bukhari, A.C.; Tusseyeva, I.; Lee, B.-G.; Kim, Y.-G. An Intelligent Real-Time Multi-Vessel Collision Risk Assessment System from VTS Viewpoint Based on Fuzzy Inference System. *Expert Syst. Appl.* **2013**, *40*, 1220–1230. [CrossRef]
21. Ohn, S.W.; Namgung, H. Interval Type-2 Fuzzy Inference System Based on Closest Point of Approach for Collision Avoidance Between Ships. *Appl. Sci.* **2020**, *10*, 3919. [CrossRef]
22. Jang, J.-S.R. ANFIS: Adaptive-network-based fuzzy inference system. *IEEE Trans. Syst. Man Cybern. Syst.* **1993**, *23*, 665–685. [CrossRef]
23. Montewka, J.; Hinz, T.; Kujala, P.; Matusiak, J. Probability modelling of vessel collisions. *Reliab. Eng. Syst. Saf.* **2010**, *95*, 573–589. [CrossRef]
24. Szłapczyński, R.; Niksa-Rynkiewicz, T. A framework of a ship domain-based near-miss detection method using mamdani neuro-fuzzy classification. *Pol. Marit. Res.* **2018**, *25*, 14–21. [CrossRef]
25. Tam, C.; Bucknall, R. Collision risk assessment for ships. *J. Mar. Sci. Technol.* **2010**, *15*, 257–270. [CrossRef]
26. Kijima, K.; Katsuno, T.; Nakiri, Y.; Furukawa, Y. On the manoeuvring performance of a ship with the parameter of loading condition. *J. Soc. Nav. Archit. Jpn.* **1990**, *168*, 141–148. [CrossRef]
27. Zhang, W.; Kopca, C.; Tang, J.; Ma, D.; Wang, Y. A systematic approach for collision risk analysis based on AIS data. *J. Navig.* **2017**, *70*, 1117–1132. [CrossRef]
28. Yasukawa, H.; Yoshimura, Y. Introduction of MMG standard method for ship maneuvering predictions. *J. Mar. Sci. Tech.* **2015**, *20*, 37–52. [CrossRef]
29. Zou, L.; Larsson, L. Numerical predictions of ship-to-ship interaction in shallow water. *Ocean Eng.* **2013**, *72*, 386–402. [CrossRef]
30. Lataire, E.; Vantorre, M.; Delefortrie, G.; Candries, M. Mathematical modelling of forces acting on ships during lightering operations. *Ocean Eng.* **2012**, *55*, 101–115. [CrossRef]
31. Yang, Y.; Eloot, K.; Delefortrie, G.; el Moctar, O. Experimental and numerical investigations of ship–ship interaction during overtaking for benchmarking. *Appl. Ocean Res.* **2025**, *154*, 104325. [CrossRef]
32. Karaboga, D.; Kaya, E. Adaptive network based fuzzy inference system (ANFIS) training approaches: A comprehensive survey. *Artif. Intell. Rev.* **2019**, *52*, 2263–2293. [CrossRef]
33. Ali, M.; Ghatol, A. A Neuro-Fuzzy Inference System for Student Modeling in Web-Based Intelligent Tutoring Systems. In Proceedings of the International Conference on Cognitive Systems, Victoria, BC, Canada, 11 December 2004.
34. Kurnaz, S.; Cetin, O.; Kaynak, O. Adaptive neuro-fuzzy inference system based autonomous flight control of unmanned air. vehicles. *Expert Syst. Appl.* **2010**, *37*, 1229–1234. [CrossRef]
35. Kosko, B. Fuzzy Associative Memory Systems. In *Fuzzy Expert Systems*; CRC Press: Boca Raton, FL, USA, 1992; pp. 135–162.
36. Kar, S.; Das, S.; Ghosh, P.K. Applications of neuro fuzzy systems: A brief review and future outline. *Appl. Soft Comput.* **2014**, *15*, 243–259. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.